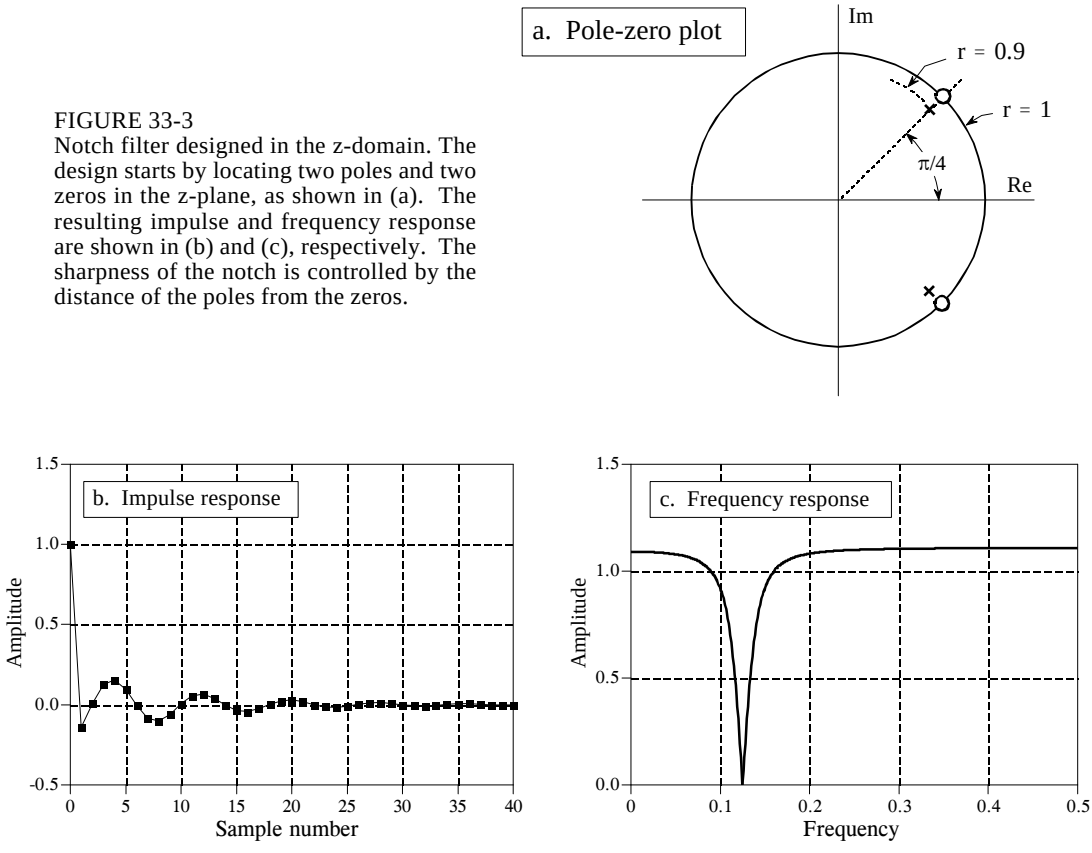

FIGURE 33-3
Notch filter designed in the z-domain. The design starts by locating two poles and two zeros in the z-plane, as shown in (a). The resulting impulse and frequency response are shown in (b) and (c), respectively. The sharpness of the notch is controlled by the distance of the poles from the zeros.

write down the transfer function in the form of Eq. 33-4, (3) rearrange the transfer function into the form of Eq. 33-3, and (4) identify the recursion coefficients needed to implement the filter. Fig. 33-3 shows the example we will use: a notch filter formed from two poles and two zeros located at

In polar form:	In rectangular form:
$z_1 = 1.00e^{j(\pi/4)}$	$z_1 = 0.7071 + j\,0.7071$
$z_2 = 1.00e^{j(-\pi/4)}$	$z_2 = 0.7071 - j\,0.7071$
$p_1 = 0.90e^{j(\pi/4)}$	$p_1 = 0.6364 + j\,0.6364$
$p_2 = 0.90e^{j(-\pi/4)}$	$p_2 = 0.6364 - j\,0.6364$

To understand why this is a notch filter, compare this pole-zero plot with Fig. 32-6, a notch filter in the s-plane. The only difference is that we are moving along the *unit circle* to find the frequency response from the z-plane, as opposed to moving along the *vertical axis* to find the frequency response from the s-plane. From the polar form of the poles and zeros, it can be seen that the notch will occur at a natural frequency of $\pi/4$, corresponding to 0.125 of the sampling rate.